

Public Goods and Common Resources

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 11** — Public Goods and Common Resources.

Read the chapter alongside these slides.

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Slide 2

Working through it

Why this chapter. Mankiw's chapter 11 covers the same two cells these slides do: public goods, which markets undersupply, and common resources, which markets overuse. It also introduces the excludable/rival classification that organizes the whole session.

What the slides add. Three things. The matrix is built here as the *answers to two questions* rather than presented as a table, so that goods can be reclassified when congestion or

technology changes. The free-rider result is derived from a two-column payoff table rather than asserted. And the commons is computed exactly — open access is shown to double the efficient fleet and destroy the resource's entire value, on numbers you can check.

How to use it. Read the chapter once before and once after. The chapter gives the vocabulary; the slides give the arithmetic that makes the vocabulary bite.

Where we are going

Where we are going

1. **Two questions to ask of any good** — the map: four kinds of goods.
2. **The free-rider problem** — why the hat comes back light.
3. **Should the government build it?** — cost-benefit analysis, and a vote that goes wrong.
4. **The tragedy of the commons** — everybody's property is nobody's.
5. **The commons, with numbers** — the overuse, counted exactly.
6. **Fixing the commons** — cap it, price it, permit it, own it.

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Slide 3

Working through it

The structure. Six parts, organized around a single two-by-two map.

Part 1 builds the map. Two yes-or-no questions — can non-payers be excluded, and does one person's use leave less — sort every good into one of four cells.

Parts 2–3 take the cell where markets undersupply. Part 2 shows *why* nobody funds a public good voluntarily; Part 3 asks whether a government should step in, and what test it should apply.

Parts 4–6 take the cell where markets overuse. Part 4 gives the mechanism in words, Part 5 computes it exactly on a fishing lake, and Part 6 tests four repairs against those same numbers.

The thread. Both failures come from one missing thing — a gate — and every repair works by rebuilding the bill that the missing gate prevented.

1. Part 1: Two Questions to Ask of Any Good

Goods without price tags

Goods without price tags

Some goods reach you through a checkout: a burrito, a haircut, a tank of gas.

Others reach you with no transaction at all: street lighting, clean harbor water, national defense, the road you drove in on.

Prices are how markets ration goods and reward suppliers.

Where no price can be charged, both jobs go undone. So: **when** does that happen, and **what** happens next?

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Slide 5

Working through it

The observation this part starts from. Some goods reach you through a transaction and some do not. A burrito requires a purchase. Street lighting does not, and neither does the harbour water quality, or national defense.

What a price actually does — two jobs, not one.

- It **ration**s: it decides who gets the good, by admitting the buyers whose willingness to pay clears it.
- It **reward**s: it pays whoever supplied the good, which is what makes supplying it worth anybody's while.

Why that matters here. Both jobs depend on a price being chargeable. For a burrito it obviously is. For street lighting it is not — and when no price can be charged, *both* jobs go

undone at once. Nobody is rationed and nobody is paid.

The two questions ahead. What follows is not a list of odd goods to memorize. It is a test, applicable to anything, that predicts whether a price can do its work.

Question one: can you be kept out?

Question one: can you be kept out?

Excludable. A good is excludable if a supplier can prevent people who have not paid from using it.

The test: **is there a gate?** Can non-payers be kept out at reasonable cost?

- **Excludable:** a burrito (hand it over or not); a streaming subscription (password).
- **Not excludable:** street lighting; fish swimming in the open ocean; national defense.

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Slide 6

Working through it

The definition. A good is **excludable** if a supplier can prevent people who have not paid from using it.

The test in plain language. *Is there a gate?* Can non-payers be kept out at reasonable cost?

Clear cases on each side. A burrito is excludable by physical handover: no payment, no burrito. A streaming subscription is excludable by technology: the password is the gate. Street lighting is not: the light falling on a paying pedestrian falls equally on the next one. National defense is not: no navy can protect only subscribers.

Why “at reasonable cost” is in the definition. Excludability is practical, not philosophical. In principle you could imagine metering street light; in practice the metering would cost more than the lighting. A gate that is technically conceivable but economically absurd is not a gate.

The consequence of that phrasing. Excludability can change without the good changing at all. Cheaper metering technology builds gates where none existed — which is exactly what happens two frames from now.

Question two: does your use leave less?

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Rival in use. A good is rival if one person's use of a unit reduces what is left for everyone else.

The test: **if I use it, is there less for you?**

- **Rival:** a burrito (eaten once); a fish (caught once); a parking space at 9 a.m.
- **Not rival:** a streamed show (your watching takes nothing from mine); street lighting; national defense.

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Slide 7

Working through it

The definition. A good is **rival in use** if one person's use of a unit reduces what is left for everyone else.

The test. *If I use it, is there less for you?*

Clear cases. A burrito, eaten once, is gone. A fish, once landed, cannot be landed again. A parking space at 9 a.m. holds one car. Against those: a streamed show, where your watching subtracts nothing from mine; street lighting; national defense.

What non-rivalry means on the cost side. If a good is non-rival, the cost of serving *one more* user is essentially zero once it exists. That is a strong statement, and it is why non-rival goods are strange: the efficient price for an extra user is zero, but a zero price cannot fund the good.

Why the two questions are independent. Rivalry is about the physics of consumption; excludability is about the feasibility of a gate. Neither implies the other, so all four combi-

nations are possible — which is precisely what the next exercise confirms, and what makes a two-by-two grid the right picture.

Two questions, four kinds of good

Two questions, four kinds of good

		Rival in use?	
		yes	no
Excludable?	yes	?	?
	no	?	?

Every good you will ever meet lands in exactly one cell — **and the cell predicts whether a market can handle it.**

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Slide 10

Working through it

Why a grid. Two independent yes-or-no questions give exactly four combinations, so the classification is complete: every good lands in one cell and no good lands in two.

How to read the axes. Across the top: is the good *rival* — does use deplete it? Down the side: is it *excludable* — is there a gate?

The claim being set up. Which cell a good lands in predicts, before you know anything else about it, whether an ordinary market will handle it well, badly, or destructively. That is a strong claim and the next four parts establish it cell by cell.

Why the grid is worth learning as a grid. The four names — private good, club good, common resource, public good — are easy to memorize and easy to misapply. Reconstructing them from the two questions takes ten seconds and never fails, whereas a memorized list fails on the first unfamiliar good. Learn the questions; the names follow.

The gated row

The gated row

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	?	?

Where there is a gate, a price can be charged — suppliers get paid, and markets form.

Club goods: the gate works even though sharing costs nothing, so access is sold by subscription.

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Slide 11

Working through it

The top row: goods with a gate.

Private goods (excludable, rival). The burrito. Everything priced in this material so far lives here. A price rations it and a price pays for it, and the market handles it without help.

Club goods (excludable, not rival). A streaming subscription. The gate works — the password keeps non-payers out — even though letting one more viewer in costs the supplier essentially nothing. Because the marginal cost is near zero but the gate holds, the natural business model is a *subscription*: sell access rather than units.

What the whole row has in common, and it is the important part. The gate lets a price exist. A price lets suppliers be paid. Paid suppliers show up. The chain runs, and the market works.

The tension inside the club-good cell. Charging for a non-rival good excludes users who value it above zero, which is inefficient — yet without the charge the good is never made. That trade-off is the entire economics of software, music and pharmaceuticals, and it is why patents and copyrights are contested rather than obvious.

The open row

The open row

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

No gate, no price — the bottom row is where markets fail.

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Slide 12

Working through it

The bottom row: no gate, and therefore no price.

Common resources (not excludable, rival). Fish in the open ocean. Anyone may take, and every taking leaves less for the rest.

Public goods (not excludable, not rival). National defense, street lighting. Nobody can be kept out, and nobody's use depletes it.

Two different injuries from the same missing gate. Recall the two jobs a price does. Without a gate:

- nobody can be *charged for supplying* the good — so it does not get supplied;
- nobody can be *charged for using it up* — so it gets used without limit.

Which injury bites depends on rivalry. If the good is non-rival there is nothing to use up, so only the first injury operates: the public good is **undersupplied**. If it is rival, there is very much something to use up, so the second injury dominates: the common resource is **overused**.

Same hole, opposite diseases. The remaining parts take one cell each.

Goods move cells

Goods move cells			
A city street, under four conditions:			
	excludable?	rival?	cell
free, empty	no	no	public good
free, jammed	no	yes	common resource
tolled, empty	yes	no	club good
tolled, jammed	yes	yes	private good

Congestion switches rivalry on. Tolling technology switches excludability on.
The cell is not a fixed property of the good.

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Working through it

The warning. A cell is not a permanent property of a good. It depends on circumstances that change, and the table makes that concrete with one piece of asphalt.

The same city street, four ways.

- **Free and empty** (3 a.m.): not excludable, not rival — your driving takes nothing from anyone. A *public good*.
- **Free and jammed** (rush hour): still no gate, but now every car slows every other. A *common resource*.
- **Tolled and empty**: a gate exists, nothing is depleted. A *club good*.
- **Tolled and jammed**: gate and depletion both. Effectively a *private good*.

What does the moving. **Congestion** switches rivalry on and off: scarcity is what makes one person's use cost another's. **Technology and law** switch excludability on and off: for most of history a city street could not be tolled, and transponders and plate cameras changed that.

Why this matters for policy. If a good's cell can be moved, moving it is itself an option. Several of the repairs in Part 6 do exactly that — they build a gate where none existed.

2. Part 2: The Free-Rider Problem

Start in the bottom-right cell

Start in the bottom-right cell

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

Not excludable, not rival: nobody can be kept out, and nobody's use leaves less. **Who would ever supply this for a living?**

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Working through it

The cell. Not excludable, not rival. Nobody can be kept out, and one supply serves everyone at once.

Why that sounds like the best cell and is the most troubled. A good that serves everybody without being used up is, from a consumption standpoint, ideal: the efficient thing is for everyone to have it. The difficulty is entirely on the supply side.

The question that exposes it. Who would supply this for a living? A seller needs paying customers. This cell, *by definition*, cannot compel anybody to become one — because non-payers cannot be excluded from the benefit.

What is coming. It is tempting to answer “people will chip in, because they want the good.” Part 2 tests that answer properly, with a specific neighborhood, a specific nuisance and a specific price, and the arithmetic turns out not to depend on how much anybody wants it.

Alder Pond

Alder Pond

A neighborhood of **120 households** around a pond. Mosquitoes, every summer.

A treatment program knocks them down for the season. Cost: **\$2,500 per season**.

Run the two questions:

- Excludable? **No** — fewer mosquitoes over the pond means fewer for every household; no gate can be built.
- Rival? **No** — one family's itch-free evening takes nothing from the next family's.

A public good, worth real money to real people.

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Working through it

The setup. 120 households around a pond. A mosquito treatment program costs **\$2,500** per season and works.

Classifying it with the two questions, which is the habit being drilled.

- **Excludable?** No. The treatment reduces mosquitoes over the whole pond. A household that refuses to pay cannot have its share of the mosquitoes put back.
- **Rival?** No. One family's mosquito-free evening subtracts nothing from the next family's.

So this is a public good — bottom-right cell.

Why the example is chosen this way. The good is unambiguously worth having: real households would pay real money for it, and Part 3 will total that up to \$3,300 against a \$2,500 cost. So the failure that follows cannot be blamed on the good being worthless. The demand is genuinely there.

The question the rest of the part answers. Can that demand be collected?

Pass the hat

Pass the hat

The neighborhood association asks every household for **\$25** toward the program ($120 \times \$25 = \$3,000 \geq \$2,500$).

Free rider. Someone who enjoys the benefit of a good without paying for it.

The program cannot exclude non-payers. So before writing a check, each household can ask: *what does my \$25 actually change?*

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Slide 19

Working through it

The natural first attempt. The neighborhood association asks each of the 120 households for \$25. If everybody gives, that is $120 \times 25 = \$3,000$ — comfortably above the \$2,500 needed.

The definition. A **free rider** is someone who enjoys the benefit of a good without paying for it.

Why free riding here requires no dishonesty. Nobody has to lie, hide, or break a promise. The treatment cannot be aimed away from a non-payer's yard. Declining to give is simply declining, and the benefit arrives anyway. This is worth stating plainly because the usual reading of "free rider" is moral, and the economics of it is not.

The question a household can now legitimately ask. Not "is this program worth \$25 to me?" — which sounds like the relevant question and is not — but "*what does my \$25 actually change?*" The next frame shows that those two questions have different answers, and that the second one is the one that governs behavior.

The free rider's arithmetic

The free rider's arithmetic

Take a household that values the season at **\$50**. With 120 households, its \$25 **does not tip** whether the program runs.

	program runs anyway	program fails anyway
chip in \$25	$50 - 25 = \$25$	-\$25
keep it	\$50	\$0

Keeping the \$25 is better in both columns.

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Working through it

The setup for the table. A household that genuinely values the season at \$50 — twice what is being asked. With 120 households, its single \$25 check does not determine whether the program reaches \$2,500. So the outcome can be treated as fixed, and the household simply asks what it gets in each case.

The four cells.

- Program runs, household chips in: benefit \$50, paid \$25, net **+\$25**.
- Program runs, household keeps its money: benefit \$50, paid nothing, net **+\$50**.
- Program fails, household chipped in: no benefit, \$25 gone, net **-\$25**.
- Program fails, household kept its money: net **\$0**.

Reading it column by column. If the program runs, \$50 beats \$25. If it fails, \$0 beats -\$25. Keeping the money is better *in both columns*.

The name for that. A **dominant** choice: one that is best regardless of what everyone else does. The household never has to predict its neighbors' behavior, which is what makes the conclusion so robust.

The hat comes back light

The hat comes back light

Every household faces the same two columns — **including the ones that value the program most.**

So the hat comes back with a fraction of \$2,500, and the program dies — *even though the neighborhood values it at well above its cost.*

Nobody cheated. Everybody reasoned. The good still died.

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Working through it

The step that makes this general. The \$50 valuation never entered the comparison. Rework the table for a household valuing the season at \$5: it compares $5 - 25 = -\$20$ against \$5, and $-\$25$ against \$0. Keeping still wins twice. Now try \$500: $500 - 25 = \$475$ against \$500, and $-\$25$ against \$0. Keeping still wins twice.

Why. The \$25 appears in both rows of each column identically, so it cancels out of the comparison. What the household values the good at changes the *size* of the numbers, never their *order*.

The consequence. Every household faces the same dominant choice — including the ones that want the program most. So the hat comes back with a few sentimental contributions and nothing close to \$2,500, and the crew is never hired.

The shape of the failure, which is the point of the frame. Nobody cheated. Nobody was irrational. A hundred and twenty reasonable people each made the individually correct choice and collectively destroyed something they all wanted. That is a much more troubling result than a story about greed, and much harder to fix by exhortation.

No gate, no revenue, no supply

No gate, no revenue, no supply

Why the market fails here, in one chain:

no exclusion $\Rightarrow$ willingness to pay never becomes revenue $\Rightarrow$ **no private supplier can cover cost**

The demand is real — \$50 households exist. The failure is that **no price can reach it**.

Public goods are **undersupplied** by markets — often supplied not at all. This is market failure by *omission*.

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Working through it

The mechanism in one chain.

no exclusion $\Rightarrow$ willingness to pay never becomes revenue $\Rightarrow$ no supplier can cover cost

Where the failure is, precisely. Not in demand. The \$50 households are real and their valuations are real. The failure is that *no price can reach* that demand, because a price requires the ability to withhold the good from non-payers, and that ability is exactly what this cell lacks.

Why this is a different failure from the ones seen before. A tax or an externality makes a market trade the *wrong quantity* — too much, too little — but the market exists and functions. Here the market may not open at all. It is market failure by **omission**: the good is simply absent.

What that implies about the remedy. Nothing done *within* the market helps, because the problem is that the market cannot form. What is needed is an institution that does not depend on voluntary payment — one that can compel contribution. That is the government, and Part 3 asks when it should.

3. Part 3: Should the Government Build It?

What taxing can do that asking cannot

What taxing can do that asking cannot

A town government holds the one power the neighborhood association lacks: **it can make paying compulsory.**

Tax every household, hire the crew — the free rider's two columns vanish, because *keep it* is no longer on the menu.

But compulsion cuts both ways. Which projects deserve it?

Cost–benefit analysis. Comparing the *total* benefit of a project to everyone with its *total* cost, to decide whether it is worth doing.

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Slide 26

Working through it

The one relevant difference. A neighborhood association asks. A government *taxes*. That is the whole of what it adds here.

Why compulsion dissolves the free-rider table. The household's dominant choice existed because "keep it" was available. Under a tax it is not, so the two-column comparison never arises. The good gets funded.

Why that is not the end of the argument. Compulsion is a serious tool and it cuts both ways. It can fund a project worth far more than it costs — and it can equally fund one worth far less, making everybody poorer while nobody could opt out. The power to compel therefore requires a test for when to use it.

The test. **Cost–benefit analysis:** compare the *total* benefit of a project, to everyone it touches, with its *total* cost.

The word doing the work is "total". A public good's benefit is spread thinly across many people, so no individual's benefit justifies the project. Summing across everyone is exactly the step the market could not perform — which is why the arithmetic looks so different

from a private purchase decision.

Write the test

Write the test

Build the program if total benefit > total cost.

Total benefit = each household's value, summed over all 120:

$$B = 40 \times \$50 + 60 \times \$20 + 20 \times \$5$$

Read the pieces:

- 40 pond-front households would pay up to \$50 each,
- 60 set back from the water, up to \$20 each,
- 20 on the far ridge, barely touched, up to \$5 each.

Cost: $C = \$2,500$.

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Working through it

Beat one: the rule, written before any arithmetic. Build if total benefit exceeds total cost.

What total benefit means for a public good. Because the good is non-rival, everybody consumes the *same* unit at once. So the value of the program is the sum of what it is worth to each household — *added*, not averaged, and not taken as the value to a typical resident.

$$B = 40 \times \$50 + 60 \times \$20 + 20 \times \$5$$

Reading the three groups. Forty households sit on the pond and would pay up to \$50 each. Sixty are set back from the water: \$20. Twenty are on the far ridge, barely troubled by mosquitoes: \$5. The spread is not decoration — it is what makes the vote in three frames' time go wrong.

The cost. $C = \$2,500$, the crew's price.

The assumption being made silently. That these values are honestly known. Hold onto that; the frame after next is where it gets its bill.

Plug the numbers in

Plug the numbers in		
group	benefit	running total
40 households $\times$ \$50	= \$2,000	\$2,000
60 households $\times$ \$20	= \$1,200	\$3,200
20 households $\times$ \$5	= \$100	\$3,300
total benefit B		\$3,300
total cost C		\$2,500
net gain $B - C$		$3,300 - 2,500 = \$800$

Verdict: build it. The town is \$800 per season richer with the program than without it.

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Slide 28

Working through it

Beat two: every row visible.

$$40 \times 50 = 2,000 \quad 60 \times 20 = 1,200 \quad 20 \times 5 = 100$$

Running totals: \$2,000, then \$3,200, then \$3,300.

The verdict.

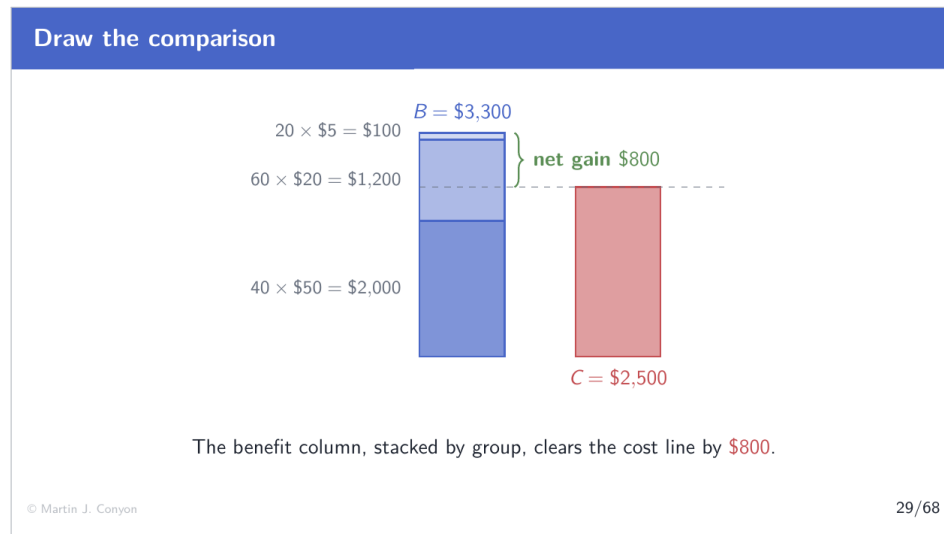
$$B - C = 3,300 - 2,500 = \$800 \text{ per season}$$

Total benefit exceeds total cost, so the test says build.

What has just been demonstrated. Two facts sit side by side and they point opposite ways. The *market* said no — the hat came back light and the program died. The *arithmetic* says yes — the town is \$800 a season richer with it than without.

Why both can be true. The market outcome reflects what people can be made to pay; the arithmetic reflects what the thing is worth. For a private good those coincide, because paying is how you get the good. For a public good they come apart, and the gap between them is exactly the space public provision occupies.

Draw the comparison



Slide 29

Working through it

Beat three: the picture. One column for benefit, stacked by group, and one for cost.

Reading the stack. The bottom block is the 40 pond-front households (\$2,000), the middle the 60 set back (\$1,200), the thin top block the 20 on the ridge (\$100). Total height: \$3,300. The cost column stands at \$2,500, and the dashed line carries that level across so the two can be compared by eye.

The brace. The benefit column clears the cost line by \$800 — the net gain, now visible rather than computed.

The second thing the picture shows, which the arithmetic buried. Look at the proportions. The bottom block alone is \$2,000 of the \$3,300: one third of the households hold nearly two thirds of the benefit. The project is efficient for the town *as a whole* while being worth very different amounts to different parts of it.

Why that asymmetry is about to matter. A cost split that ignores it will charge the ridge households more than the program is worth to them — and they vote.

Where would the numbers come from?

Where would the numbers come from?

The test needed each household's true value. But a public good has **no market and no price** — there is nothing to observe.

So the town must *ask*. And asking invites strategy:

- If your stated value sets your **tax** → understate it.
- If it just makes the project likelier → **overstate** it.

The self-interest that emptied the hat also bends the survey.

In practice: benefits are *estimates* — cost–benefit analysis is honest arithmetic on uncertain inputs.

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Slide 30

Working through it

Paying the bill for the assumption. The test consumed one number per household: their true willingness to pay. Where would a town actually get it?

Why the usual source is unavailable. For a private good, the market reveals valuations: paying \$4 for a coffee is a confession that it was worth at least \$4 to you. A public good has no register and no purchases, so there is nothing to observe.

So the town must ask — and asking has consequences. The moment a stated value affects an outcome, answers become strategic:

- If your stated value determines your **tax**, understating it lowers your bill while the program still goes ahead.
- If it merely makes the project **likelier**, overstating it is free and helps.

The same force, in a new costume. This is the self-interest that emptied the hat, now bending a survey. It is not a separate problem; it is the same one.

What follows for practice. Real cost–benefit analysis is honest arithmetic performed on estimated inputs. Its verdicts deserve to be taken seriously and not treated as measurements.

Put it to a vote

Put it to a vote

Fund it the obvious way: **equal shares**, $2,500/120 \approx \$20.83$ per household.
Referendum:

group	value	tax	net	vote
40 households	\$50	\$20.83	$50 - 20.83 = +\$29.17$	yes
60 households	\$20	\$20.83	$20 - 20.83 = -\$0.83$	no
20 households	\$5	\$20.83	$5 - 20.83 = -\$15.83$	no
result				no wins, 80–40

An efficient project loses two to one.

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Slide 31

Working through it

The obvious funding rule. Split the cost equally: $2,500/120 \approx \$20.83$ per household.

Each group's private arithmetic.

- 40 pond-front: $50 - 20.83 = +\$29.17$ — **yes**.
- 60 set back: $20 - 20.83 = -\$0.83$ — **no**, by 83 cents.
- 20 on the ridge: $5 - 20.83 = -\$15.83$ — **no**, emphatically.

The result. 80 against, 40 in favor. **An efficient project loses two to one.**

What has and has not gone wrong. Nothing was miscounted. The town really is \$800 richer with the program. And no voter behaved badly: each correctly compared their own benefit with their own bill.

The lesson, stated carefully. A positive *sum* does not imply a positive *majority*. Total surplus is a statement about the aggregate; voting is a statement about the distribution of individual nets. When benefits are concentrated and costs are spread evenly, the two can point in opposite directions — and the next frame shows that the split, not the project, is what must change.

Change the split, change the vote

Change the split, change the vote

Same project, same \$2,500 — **shaped to benefit** instead of split equally:

group	value	tax	revenue	net
40 households	\$50	\$38	$40 \times 38 = \$1,520$	+\$12 each
60 households	\$20	\$15	$60 \times 15 = \$900$	+\$5 each
20 households	\$5	\$4	$20 \times 4 = \$80$	+\$1 each
total			\$2,500	yes, 120–0

The obstacle was the split, not the project. But shaping taxes to benefit requires knowing the values — *the very numbers people hide*.

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Working through it

Same project, same \$2,500, different shares. Charge \$38 to the pond-front households, \$15 to those set back, \$4 on the ridge.

Check the revenue.

$$40 \times 38 = 1,520 \quad 60 \times 15 = 900 \quad 20 \times 4 = 80$$

$1,520 + 900 + 80 = \$2,500$ exactly. The program is fully funded.

Check each group's net. $50 - 38 = +\$12$; $20 - 15 = +\$5$; $5 - 4 = +\$1$. **Every household is better off with the program than without it**, so the referendum passes 120 to 0.

The conclusion. The obstacle was never the project. It was the split. Any project with $B > C$ has *some* allocation of the cost under which everyone gains — because the total gain is positive, there is enough to go round.

The catch, and it is a real one. Shaping taxes to benefit requires knowing each group's true value — which the previous frame established are the very numbers self-interest hides. Efficient projects are feasible only under splits that require unknowable numbers. That knot is the honest difficulty of public provision, and it is why these arguments fill town meetings.

Public goods in practice

Public goods in practice

National defense — the classic pure case: roughly \$900 billion of U.S. spending in 2024, covering every resident, excluding none.

Basic research — an idea, once published, is non-rival and non-excludable: a theorem protects no one who funds it.

- A **patent** builds a legal gate around an *applied* idea → moves invention up to the club/private row → private R&D happens.
- Basic science has no such gate — so it runs on public funding and philanthropy.

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Working through it

National defense: the pure case. Roughly \$900 billion of U.S. spending in 2024, covering every resident and excluding none. There is no way to defend the airspace over paying households only, so the free-rider problem is total and the good is entirely tax-funded.

Basic research: the instructive case. An idea, once published, is non-rival — your using a theorem leaves all of it for me — and non-excludable, since a published result cannot be fenced. So it sits squarely in the public-good cell, and private funding for it is correspondingly scarce.

What a patent is, in the language of the matrix. A **legal gate** built around an *applied* idea. It converts a non-excludable good into an excludable one, moving it up into the club/private row — and that is precisely why private firms will fund drug development.

Why the gate stops where it does. Deep results cannot be patented: you cannot own a theorem or a physical law. So the layer underneath applied invention has no gate available and no private funding model, which is why basic science runs on public money and philanthropy. The pattern is not an accident of policy; it follows from where a gate can and cannot be built.

The cell describes the good, not the supplier

The cell describes the good, not the supplier

public goods, privately supplied	private goods, publicly supplied
Wikipedia (donations, volunteers)	a public-school seat
ad-funded broadcast radio	(rival <i>and</i> excludable)
open-source software	a city bus ride

Public good is a statement about **excludability and rivalry** — never about who happens to provide it.

Private provision of public goods is real but fragile: it leans on donations, advertising, or volunteers — **not on a price for the good itself**.

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The confusion worth killing. “Public good” does not mean “government-provided good”. The matrix classifies goods by excludability and rivalry, and says nothing whatever about who supplies them.

Public goods, privately supplied. Wikipedia is non-rival and non-excludable and is supplied by donations and volunteers. Ad-funded broadcast radio transmits a public good and finances it by selling listeners’ attention. Open-source software runs much of the internet.

Private goods, publicly supplied. A seat in a public school is rival (one child occupies it) and excludable (enrolment can be refused). A city bus ride is the same. Both are private goods that a government happens to provide, usually for distributional reasons rather than because a market could not.

What every private-supply example has in common. Look at how each is financed: donations, advertising, volunteered labor. Never a price charged for the good itself — because that is exactly what non-excludability forbids.

So the qualification. Private provision of public goods is real, and it is fragile. It depends on sideways revenue, and sideways revenue is neither reliable nor proportioned to how much the good is worth.

4. Part 4: The Tragedy of the Commons

The dangerous cell

The dangerous cell

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

Common resource. A good that is rival in use but not excludable: anyone may take, and every taking leaves less.

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Working through it

The definition. A **common resource** is rival in use but not excludable: anyone may take, and every taking leaves less.

Why the combination is destructive. Take the two halves in turn.

- **Rivalry** means use has a genuine cost — what I take is gone from what you can take. There is something real to be lost.
- **Non-excludability** means nobody can be made to pay that cost, or stopped at the gate, because there is no gate.

So a real cost exists and no mechanism exists for charging it.

Contrast with the cell just left. A public good is non-rival, so there is nothing to deplete; its failure is that nobody will *supply* it. A common resource has plenty to deplete and no way to limit the depleting; its failure is that everybody *takes* it.

Starvation against devouring. Two failures from one missing gate, running in opposite

directions. That symmetry is worth holding, because the repairs in Part 6 are the mirror image of the tax-and-provide remedy in Part 3.

An old story

An old story

A village pasture, open to every herder.

- Each animal added is **pure gain to its owner** — the grass it eats is free.
- The grass it eats is **loss to every herd on the pasture** — shared by all, charged to none.
- So herds grow past what the pasture can renew... and the pasture dies. Everyone's animals starve.

The name comes from Garrett Hardin's 1968 essay "The Tragedy of the Commons" (*Science*).

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Working through it

The story. A village pasture open to every herder.

The herder's private arithmetic, item by item.

- An added animal is **pure gain** to its owner: its milk, its meat, its market price all belong to the herder.
- The grass it eats is a **loss spread across every herd** on the pasture — including, but only fractionally, the herder's own.

Why that produces the outcome. With, say, twenty herders, an added animal imposes roughly nineteen-twentieths of its grazing cost on *other people*. The owner weighs the whole gain against a twentieth of the cost, and the answer is add another. Every herder faces the same tilted comparison, so every herd grows.

The end state. Herds expand past what the pasture can renew, the grass is destroyed, and every animal on it starves — including the animals of the herders who saw it coming.

The citation. The name comes from Garrett Hardin’s 1968 essay “The Tragedy of the Commons” in *Science*. Fifty years of subsequent work, surveyed by Frischmann, Marciano and Ramello (2019), has refined *when* the tragedy strikes — and Part 6 covers the cases where communities dodge it.

The diagnosis: a cost nobody pays

The diagnosis: a cost nobody pays

When my use of a good imposes a cost on you that I never pay, **I use too much of it.**
That is the whole mechanism.

The herder counts *all* of the added animal’s value — but bears only a *sliver* of the grass it destroys.

Public goods: too little gets made.
Common resources: too much gets taken.

Two cells, one missing gate — failing in opposite directions.

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Working through it

The mechanism in one sentence. When my use of something imposes a cost on you that I never pay, I use too much of it.

Why “too much” is precise rather than rhetorical. A user weighs their *full* private gain against the *fraction* of the true cost that lands on them. Any use whose full social cost exceeds its gain, but whose private share of that cost does not, gets undertaken. Those are exactly the uses that should not happen, and there is a definite number of them — Part 5 counts them.

Applied to the herder. Full value of the animal, one herd’s sliver of the grass damage. The rest lands on neighbors.

Why overuse is an equilibrium, not an accident. Every herder faces the same tilted comparison simultaneously. Nobody has to coordinate, and nobody deviating would help — a

restrained herder simply leaves grass for the others. Restraint is not merely unrewarded; it is futile.

The pair, set side by side. Public goods: too little gets made, because nobody can charge for supplying. Common resources: too much gets taken, because nobody is charged for using up. One missing gate, two diseases.

Commons all around us

Commons all around us

- **Ocean fisheries** — the northern cod off Newfoundland collapsed; Canada closed the fishery in 1992, and it has never fully returned.
- **Groundwater** — the Ogallala aquifer, drawn down faster than rain refills it, well by private well.
- **Rush-hour roads** — every driver slows every other driver, and no one is billed for it.
- **Antibiotics** — each course of use erodes a shared pool of effectiveness that no prescriber pays for.

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Working through it

Why these four. Each is unmistakably rival and unmistakably ungated, and between them they span four centuries and four industries.

Ocean fisheries. The northern cod off Newfoundland fed Europe for five hundred years. Open access plus industrial trawlers crashed the stock; Canada closed the fishery in 1992, roughly thirty thousand people lost their livelihoods, and the stock has never fully recovered. Note what makes it a commons rather than bad luck: no vessel bore the cost its catch imposed on every other vessel's future catch.

Groundwater. The Ogallala aquifer under the Great Plains is drawn down faster than rainfall refills it, one privately rational well at a time. Each farmer's pumping lowers the water table for every neighbor.

Rush-hour roads. Every driver slows every other driver and is billed nothing for it. This is the commons most people experience daily.

Antibiotics. Each course prescribed anywhere erodes a shared, finite pool of effectiveness. No prescriber pays for the erosion, and the resource — the usefulness of a drug class — is genuinely depletable.

The common structure. Different centuries, same tilted arithmetic.

No villains required

No villains required

The tragedy needs **no greed, no malice, no fools** — only ordinary self-interest plus open access.

Blaming the users misreads the mechanism: *anyone* in their position faces the same arithmetic.

So the repair is never “find better people.”
It is: change the arithmetic.

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Working through it

The claim. The tragedy needs no greed, no malice and no stupidity — only ordinary self-interest plus open access.

Why that is worth insisting on. It is tempting to read the cod collapse or a drained aquifer as a story about bad actors. But every user in the model behaves exactly as a careful, decent person would: they compare their own gain with their own cost and act accordingly. Replace all of them with more virtuous people facing the same rules and the outcome is unchanged.

The test of that claim. Ask what a restrained user achieves. On an open commons, one herder who holds back leaves grass that the others take. The restraint transfers the resource

rather than saving it. So restraint is not just unrewarded — it is ineffective, which is why moral appeals fail here in a way they do not always fail elsewhere.

What follows for repair, and it is the practical payoff. Campaigns to shame users into restraint are fighting the arithmetic, and the arithmetic usually wins. Every fix in Part 6 — caps, prices, permits, ownership, community rules — works instead by **changing the arithmetic** so that restraint becomes the privately sensible choice.

5. Part 5: The Commons, with Numbers

Birch Lake: write the equation

Birch Lake: write the equation

A lake with good fishing, open to all. A boat-day costs **\$400** (fuel, gear, wages). Boats crowd each other:

$$\text{catch per boat} = 1200 - 100b \quad (b = \text{boats on the water})$$

Read the coefficients:

- **1200**: what one boat alone lands in a day (\$ of fish).
- **100**: what every boat loses when one more launches — the crowding term.

Total catch = boats $\times$ catch per boat: $T(b) = b(1200 - 100b)$.

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Working through it

Beat one: the model, in one line. A lake open to all. A boat-day costs **\$400** — fuel, gear, crew wages, all in. Boats crowd each other:

$$\text{catch per boat} = 1200 - 100b$$

where b is the number of boats on the water, and catch is measured in dollars of fish.

Reading both coefficients.

- **1200**: what a single boat lands on an empty lake.
- **100**: the crowding term. Every boat already fishing loses \$100 of catch the moment one more launches. This one number is the entire externality, and every result below traces back to it.

Total catch. Boats times catch per boat:

$$T(b) = b(1200 - 100b)$$

Why this is enough. There is no calculus anywhere in this part. One linear equation, tabulated, produces the efficient fleet, the open-access fleet, and the exact dollar value destroyed — all by arithmetic you can check row by row.

Plug the numbers in

Plug the numbers in			
boats b	catch per boat $(1200 - 100b)$	total catch $(b \times \text{per boat})$	extra from last boat
1	$1200 - 100 = 1100$	$1 \times 1100 = 1100$	—
2	$1200 - 200 = 1000$	$2 \times 1000 = 2000$	$2000 - 1100 = 900$
3	$1200 - 300 = 900$	$3 \times 900 = 2700$	$2700 - 2000 = 700$
4	$1200 - 400 = 800$	$4 \times 800 = 3200$	$3200 - 2700 = 500$
5	$1200 - 500 = 700$	$5 \times 700 = 3500$	$3500 - 3200 = 300$
6	$1200 - 600 = 600$	$6 \times 600 = 3600$	$3600 - 3500 = 100$
7	$1200 - 700 = 500$	$7 \times 500 = 3500$	$3500 - 3600 = -100$
8	$1200 - 800 = 400$	$8 \times 400 = 3200$	$3200 - 3500 = -300$
9	$1200 - 900 = 300$	$9 \times 300 = 2700$	$2700 - 3200 = -500$

Two columns to watch: what a boat **catches** (per boat), and what a boat **adds** (extra). **They are not the same number.**

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Working through it

Beat two: build the table.

Column 2, catch per boat. Substitute into $1200 - 100b$: at $b = 1$, \$1,100; at $b = 2$, \$1,000; falling by \$100 each row.

Column 3, total catch. Boats times column 2. At $b = 4$: $4 \times 800 = 3,200$. Read the column down: it climbs to **\$3,600 at six boats** and then falls. Crowding eventually beats headcount.

Column 4, extra from the last boat. Differences of column 3. The second boat adds $2,000 - 1,100 = \$900$. The fifth adds $3,500 - 3,200 = \$300$. From the seventh on it is *negative*: the seventh boat adds $3,500 - 3,600 = -\$100$, actually shrinking the total haul.

The two columns to keep separate, because the whole part turns on it. What a boat **catches** (column 2) and what a boat **adds** (column 4) are different numbers. At five boats a boat catches \$700 and adds \$300.

Why they differ. A boat's catch includes fish that other boats would otherwise have taken. Its *addition* counts only genuinely new fish. The gap between the two columns is the crowding it imposes.

Route one: the net column

Route one: the net column			
boats b	total catch	cost ($400b$)	net (catch – cost)
1	1100	400	$1100 - 400 = 700$
2	2000	800	$2000 - 800 = 1200$
3	2700	1200	$2700 - 1200 = 1500$
4	3200	1600	$3200 - 1600 = 1600$
5	3500	2000	$3500 - 2000 = 1500$
6	3600	2400	$3600 - 2400 = 1200$
7	3500	2800	$3500 - 2800 = 700$
8	3200	3200	$3200 - 3200 = 0$
9	2700	3600	$2700 - 3600 = -900$

The net column peaks at **4 boats: \$1,600 a day.**

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Working through it

The planner's question. How many boats should fish, counting everybody's fish and everybody's costs?

The calculation. Net = total catch – 400*b*, row by row:

$$b = 1 : 700 \quad b = 2 : 1,200 \quad b = 3 : 1,500 \quad \mathbf{b = 4 : 1,600} \quad b = 5 : 1,500$$

and onward down to \$0 at eight boats and –\$900 at nine.

The answer. The net column peaks at **four boats**, worth **\$1,600 a day**. That is what the lake is genuinely worth when it is used well.

Two features worth noting for later. First, net is still \$1,200 at six boats, where *total catch* peaks — so maximizing the catch is not the same as maximizing the value, because catching costs money. Second, net hits exactly zero at eight boats. That number is not a coincidence, and the frame after next explains why open access lands on precisely it.

Route one, stated as a method. Tabulate net and take the peak. It always works, and it requires no theory at all — which is exactly why it is worth doing before the cleverer route.

Route two: stop when the extra stops covering

Route two: stop when the extra stops covering

Add a boat only while the **extra catch** covers the **\$400** it costs to launch:

boat	extra catch	vs. \$400	verdict
3rd	\$700	700 > 400	launch
4th	\$500	500 > 400	launch
5th	\$300	300 < 400	stay ashore

Stop at **4 boats** — the same answer, and *it must be*: the net column climbs exactly as long as each boat adds more than it costs.

Two routes, one answer: the check that catches errors.

Working through it

The method. Add boats one at a time, and launch a boat only while the *extra catch* it brings covers the \$400 it costs.

Applying it.

- 3rd boat: adds \$700 against \$400 — launch.
- 4th boat: adds \$500 against \$400 — launch.
- 5th boat: adds \$300 against \$400 — **stay ashore**.

Stop at four boats.

Why it must give the same answer as route one. The net column rises exactly as long as each added boat brings in more than it costs, and falls once it does not. So “the peak of net” and “the last boat worth adding” are the same fact read two ways — not a coincidence to be checked but an identity.

Why run both anyway. Because the identity is what makes the check powerful. Two routes that *must* agree, computed independently, will disagree only if there is an arithmetic error. The disagreement is an alarm, and a free one.

Which column route two uses. The *extra* column — what a boat adds. Keep that in view: the very next frame runs the identical procedure on the *other* column, and gets a different fleet.

What open access actually does

What open access actually does

Nobody asks what their boat *adds*. Each skipper asks: “**will my boat’s catch cover my \$400?**” — the *per-boat* column:

boat	catch per boat	vs. \$400	decision
6th	\$600	$600 > 400$	launch
7th	\$500	$500 > 400$	launch
8th	\$400	$400 = 400$	launch (breaks even)
9th	\$300	$300 < 400$	stays ashore

Entry stops only at **8 boats** — twice the number the lake can profitably carry.

Working through it

The change of question. No skipper asks what their boat *adds* to the total — that number falls on other people's decks and appears nowhere in their own accounts. Each asks only: *will my catch cover my \$400?*

Which is the per-boat column.

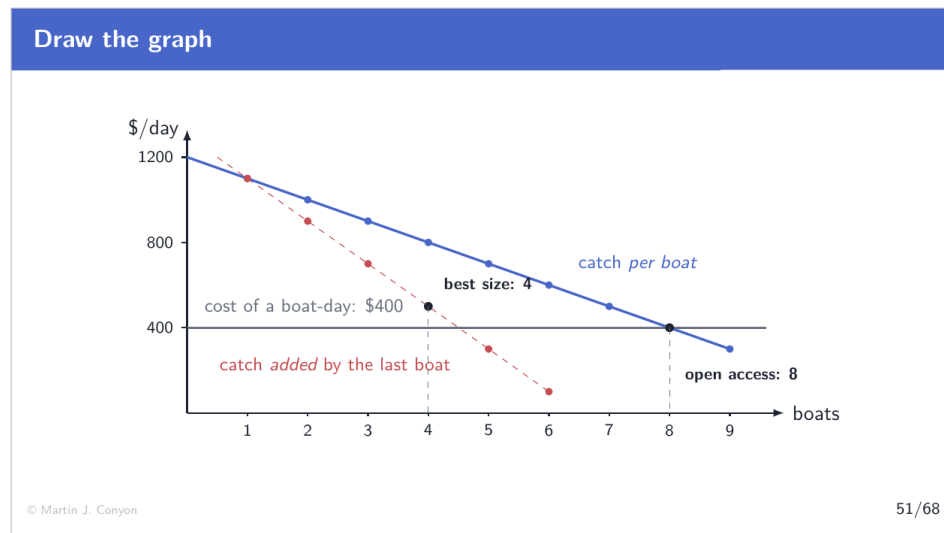
- 6th boat: catches \$600 against \$400 — launch.
- 7th boat: catches \$500 against \$400 — launch.
- 8th boat: catches \$400 against \$400 — exactly breaks even, so it launches (indifferent, and in practice it enters).
- 9th boat: catches \$300 against \$400 — stays ashore.

Entry stops at eight boats — twice the four the lake can profitably carry.

What did and did not change between the two frames. Same table. Same skippers. Same \$400 cost. Same rule of thumb — launch while it covers cost. The *only* difference is which column the decision reads: what a boat adds, or what a boat catches.

Why entry stops precisely where net hits zero. Entry continues until catch per boat equals cost, which is the condition that each boat exactly breaks even. If every boat breaks even, the fleet's total profit is zero — which is the \$0 net at eight boats spotted two frames ago.

Draw the graph



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Working through it

Beat four: dots from the table, lines from the function.

The blue line is catch per boat, $1200 - 100b$, sliding down as the lake fills. Its dots are column 2 of the table.

The red dots are the extra catch each successive boat adds. Notice they fall *twice as fast* — by \$200 per boat rather than \$100. The reason is worth stating: each entrant not only catches less itself, it also removes \$100 from every boat already out there. Both effects hit the addition; only the first hits the average.

The grey horizontal is the \$400 cost of a boat-day.

The two crossings, which are the two answers.

- The *red* dots cross cost at four boats: the lake's best size.
- The *blue* line crosses cost at eight boats: where open access lands.

The tragedy as a picture. It is the horizontal gap between the two crossings — four boats of pure waste. They launch because skippers read the blue line when the lake needed them to read the red dots, and nothing in an open fishery makes anybody read the red dots.

The lake's rent, eaten alive

The lake's rent, eaten alive		
	best size (4 boats)	open access (8 boats)
total catch	\$3,200	\$3,200
cost of boats	$4 \times 400 = \$1,600$	$8 \times 400 = \$3,200$
net, per day	\$1,600	\$0

Eight boats burn \$3,200 to land what four boats land for \$1,600.

Open access destroys every dollar the lake was worth.

Each boat exactly breaks even — the lake's *rent* is fully dissipated.

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Working through it

The comparison, and the striking row is the first one.

- Four boats: total catch **\$3,200**; cost $4 \times 400 = \$1,600$; net **\$1,600**.
- Eight boats: total catch **\$3,200**; cost $8 \times 400 = \$3,200$; net **\$0**.

Check the catch figures against the first table. At $b = 4$, $4 \times 800 = 3,200$. At $b = 8$, $8 \times 400 = 3,200$. The two fleets land *the same fish*.

So what did the extra four boats buy? Nothing. They doubled the fuel, the gear and the crews, and landed not one extra dollar of fish. The \$1,600 a day the lake was genuinely worth has been burned up as duplicated effort.

The term for it. Economists say the lake's **rent** — the value the resource itself generates, over and above the cost of working it — has been *fully dissipated*. Every boat exactly breaks even, so nobody is even profiting from the destruction.

The result to carry. An unowned resource of real value can be reduced to precisely zero net value, with every participant behaving reasonably and nobody gaining. That is a stronger statement than “overuse” and it is what makes the commons worth a whole part.

Catch is not contribution

Catch is not contribution

Why does the 5th skipper launch, when the lake wants 4?

the skipper's ledger	
my catch ($b = 5$)	+\$700
my cost	-\$400
my profit	+\$300 $\Rightarrow$ launch

the lake's ledger	
taken off the 4 boats out (4×100)	-\$400
genuinely new fish ($700 - 400$)	+\$300
cost of the 5th boat	-\$400
the lake's change	-\$100

Of the \$700 caught, \$400 was merely *moved* — taken from other decks.

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Working through it

The question. Why does the fifth skipper launch when the lake wants four?

The skipper's ledger. At $b = 5$, catch per boat is \$700. Against \$400 of cost, that is \$300 of profit. Launch — and the decision is correct, given what the skipper faces.

The lake's ledger, for the same boat.

- **Taken from the four boats already out:** $4 \times \$100 = -\400 . This is the crowding coefficient doing its work.
- **Genuinely new fish:** $700 - 400 = +\$300$.
- **Cost of the fifth boat:** $-\$400$.

Net change to the lake: $300 - 400 = -\$100$.

The sentence that names the mechanism. **Catch is not contribution.** Of the \$700 the fifth boat lands, \$400 is merely *moved* — fish that would have been caught anyway, taken off other decks. Only \$300 is new.

And the mover is never billed. That unbilled \$400 is exactly the “cost nobody pays” from

the pasture story, now wearing a number. Every repair in Part 6 is an attempt to put that \$400 on the fifth skipper's own ledger.

6. Part 6: Fixing the Commons

One problem, four fixes

One problem, four fixes

The broken arithmetic, in one line: **takers never pay the cost their taking imposes on the other users.**

Every working repair re-attaches that cost. Four families:

1. **Cap it** limit use directly
2. **Price it** charge for each use
3. **Permit it** cap, then let the caps trade
4. **Own it** give someone the whole

Each one gets tested on **Birch Lake** — does it get us back to 4 boats and \$1,600 a day?

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Working through it

The disease, stated precisely. Takers never pay the cost their taking imposes on the other users — the unbilled \$400 in the fifth skipper's ledger.

What every working repair therefore does. Re-attach that cost to the person making the decision. Everything else is implementation.

The four families.

1. **Cap it** — limit use directly.
2. **Price it** — charge for each use.
3. **Permit it** — cap, then let the caps trade.
4. **Own it** — give the whole resource to someone.

How they differ, and this is what to watch for. Not in whether they work — all four can — but in three practical dimensions: what the authority must *know*, what must be *policed*, and who ends up with the resource's *value*.

The test each must pass. Birch Lake, back to four boats and \$1,600 a day. Using one worked example for all four makes the comparison honest: any difference in the verdict is a difference in the instrument, not in the case.

Cap it

Cap it

License exactly 4 boats. Fleet: 4. Net: back to \$1,600 a day. *It works.*

What it demands:

- The regulator must **know the lake's table** — cap at 6 and most of the waste survives; cap at 2 and value is thrown away.
- Someone must **police** the water.
- Someone must decide **who** gets the 4 licenses — queue? lottery? connections?

The real world's version: fishing seasons, catch limits, hunting licenses, well-drilling permits.

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Working through it

The instrument. License exactly four boats. Fleet: four. Net: \$1,600 a day. **It works** — the outcome is exactly the efficient one.

What it demanded, item by item.

Knowledge. The regulator had to know the lake's table. Cap at six and most of the waste survives; cap at two and value is thrown away in the other direction. The cap is only as good as the number chosen, and choosing it requires precisely the information a distant authority is least likely to have.

Enforcement. At four boats, a fifth entrant privately clears \$300 a day. The incentive to cheat is created *by the cap's own success* — the better it works, the more profitable

violating it becomes. So policing is not incidental; it is permanent.

Allocation. Somebody must decide *who* gets the four licences. A queue, a lottery, an auction, or the mayor's fishing friends. The cap fixes the quantity and says nothing about the recipients, so the choice is made administratively — and it is worth a great deal of money.

Where you meet it. Fishing seasons, catch limits, hunting licences, well-drilling permits. All inherit all three demands.

Price it

Price it

Charge a fee per boat-day — say \$350. A boat-day now costs $400 + 350 = \$750$ to the skipper.

boat	catch per boat	vs. \$750	decision
4th	\$800	$800 > 750$	launch
5th	\$700	$700 < 750$	stays ashore

Entry stops at **4 boats** — the fee makes each skipper *feel* the crowding cost they used to dump on the fleet.

Any fee over \$300, up to \$400, lands on the same 4 boats.

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Working through it

The instrument. Leave entry open, but charge a fee per boat-day — say \$350. The skipper's all-in cost becomes $400 + 350 = \$750$.

Rerunning the open-access logic at the new cost. Skippers still read the per-boat column; only the number they compare against has changed.

- 4th boat: catches \$800 against \$750 — launch.
- 5th boat: catches \$700 against \$750 — **stays ashore**.

Entry stops at four boats — the efficient fleet, reached by the skippers' own self-interest with nobody licensed or turned away.

What the fee is doing. It makes each skipper *feel* the crowding cost they used to dump on the fleet. The fifth boat's unbilled \$400 is now approximated by a \$350 bill the skipper actually pays.

The range, which matters practically. The answer is not knife-edge. Any fee above \$300 keeps the fifth boat ashore ($700 < 400 + f$ requires $f > 300$), and any fee below \$400 keeps the fourth boat fishing ($800 > 400 + f$ requires $f < 400$). **Any fee strictly between \$300 and \$400** lands entry on the same four boats — so the regulator needs a range, not a bullseye.

Price it: where the value goes

Price it: where the value goes

With 4 boats and the \$350 fee, the \$1,600 a day is **preserved** — and split:

- boats keep $4 \times (800 - 750) = \$200$,
- the town collects $4 \times 350 = \$1,400$ — available to cut other taxes.

Congestion pricing is this fix on asphalt.

The IGM panel of leading economists, on peak-hour road pricing: tolls at peak times, with the proceeds used to lower other taxes, “would make citizens on average better off.”

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Working through it

Following the money. A fee that stops the waste has to send the recovered value somewhere, and it is worth tracking where.

The split at four boats and a \$350 fee.

- Boats keep $4 \times (800 - 750) = \$200$ a day.

- The town collects $4 \times 350 = \$1,400$ a day.

Total: \$1,600 — the lake's full value, restored and accounted for.

What changed and what did not. The *quantity* of value is the same as under the cap. What differs is its destination: under open access the \$1,600 was burned as duplicated fuel and crews; under the fee it is *collected*, and available to cut other taxes or fund something else. Nothing was destroyed and nothing was created — it was rescued.

The real-world version. Congestion pricing is this fix on asphalt: a peak-hour charge that makes each driver feel the delay they impose on everyone behind them.

The evidence on it. The IGM panel of leading economists, surveyed across the political spectrum, agreed near-unanimously that peak-hour road tolls, with the proceeds used to lower other taxes, “would make citizens on average better off.” Near-unanimity on that panel is rare and worth noticing.

Permit it

Permit it

Cap at 4 boat-days — then let the permits trade.

What is a permit worth? Fishing with one clears $800 - 400 = \$400$ a day — so permits sell for **about \$400**, and end up with *the skippers who fish best*.

The cap sets **how much**; the trading decides **who** — the regulator no longer has to.

Real fisheries run on this: individual transferable quotas in New Zealand and Iceland since the 1980s. Alaska's halibut season had shrunk to a 2–3 day derby; under tradable quotas (1995) it runs about 8 months — same fish, no race.

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Working through it

The instrument. Cap the lake at four boat-days — then issue the cap as *permits* and let them trade.

What a permit is worth. At a four-boat fleet a boat catches \$800 and spends \$400, so fishing with a permit clears \$400 a day. Competition for permits therefore bids their price to roughly **\$400**.

What the permit price accomplishes, quietly. A skipper who can fish efficiently values a permit above \$400 and buys; one who cannot values it below and sells. So the four boats on the water end up being *the right four*, selected by willingness to pay rather than by an official's judgement.

The division of labor. The cap decides **how much**. The trading decides **who**. That is the crucial improvement on a plain cap, which had to answer both questions administratively.

This is not a blackboard scheme. New Zealand and Iceland have run their fisheries on individual transferable quotas since the 1980s. Alaska's halibut fishery is the clearest before-and-after: under open access the season had compressed into a two-or-three-day derby, with boats putting to sea in storms rather than lose their window. Under tradable quotas from 1995 the same catch comes ashore over roughly eight months, calmly. Same fish, no race.

Own it

Own it

Give the lake one owner.

The owner's profit *is* the net column — total catch minus boat costs. Reading it, the owner runs **4 boats** and pockets **\$1,600 a day**. No regulator, no table of fees.

Ownership makes the crowding cost land on the chooser.

A fifth boat would move \$400 between the owner's own decks and add \$300 of fish for \$400 of cost — **the owner simply declines**.

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Working through it

The instrument. Give the lake a single owner. No regulator, no fee schedule, no licensing

office.

Why it works immediately. The owner's profit *is* the net column — total catch minus total boat costs. Any owner reading that column runs four boats and pockets \$1,600 a day, because that is what maximizes their own money.

What happened to the externality. It did not get taxed or capped; it got **internalized**. When the owner considers a fifth boat, the \$400 it strips from the other four boats is stripped from *the owner's own decks*. The private ledger and the lake's ledger become the same document:

$$+\$300 \text{ new fish} - \$400 \text{ cost} = -\$100$$

so the owner declines. Nobody had to make them.

The general principle. Ownership makes the crowding cost land on the chooser. That is why a pasture divided into fenced family plots stops being a tragedy: each family now eats its own overgrazing, and the arithmetic that drove the tragedy simply no longer applies.

What it requires. A boundary that can be drawn and defended, and a legal system that recognizes it. Where those exist, ownership is the cheapest fix on the list. Where they do not — the open ocean, the atmosphere — it is unavailable, which is why the other three exist.

Own it, applied: elephants

Own it, applied: elephants

Wildlife is a commons wherever it is **unowned**: a poached elephant is pure gain to the poacher; the loss falls on everyone — and on no one in particular.

- **Kenya**, 1973–1989: elephants unowned, hunting banned — the herd fell from roughly **167,000 to 16,000**. Bans without owners left no one whose loss a dead elephant was.
- **Southern Africa**: Zimbabwe's CAMPFIRE (from 1989) and its neighbors gave local communities rights to wildlife and its revenues — elephants became an asset to protect. The region now holds **the continent's largest herds**.

Where someone owns the elephant's value, someone guards it.

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Working through it

Wildlife as a commons. An unowned elephant is rival (poached once) and non-excludable (nobody can be kept from the herd). To a poacher it is pure gain; the loss falls on everyone in general and nobody in particular.

Kenya, 1973–1989. Elephants unowned, hunting banned, enforcement attempted. The herd fell from roughly **167,000 to 16,000**. The lesson is not that bans are pointless but that a ban creates no *owner*: there is still nobody whose personal loss a dead elephant is, and so nobody with a standing incentive to guard it.

Southern Africa, from 1989. Zimbabwe’s CAMPFIRE programme and comparable schemes in neighboring countries granted local communities enforceable rights to wildlife and to its revenues — tourism, regulated hunting fees. An elephant became an asset the village protects the way a rancher protects cattle. The region now holds the continent’s largest herds.

The mechanism, stated carefully. Ownership does not make people care about elephants. It recruits self-interest to the elephant’s side, which is more reliable than caring. The animal is worth more alive to the person in a position to protect it.

The honest qualification. The comparison is not a controlled experiment — these countries differ in many ways. But the direction of the result matches the theory, and the theory predicted it in advance.

Own it together: Ostrom’s commons

Own it together: Ostrom’s commons

Between the market and the state: **communities that govern their own commons** — and have, for centuries.

- Swiss alpine pastures: village grazing rules running since the Middle Ages.
- Maine lobster harbors: local territories and self-enforced limits.
- Spanish *huertas*: irrigation water shared by farmer-run courts for a thousand years.

Elinor Ostrom (*Governing the Commons*, 1990; Nobel 2009): what the survivors share — **clear boundaries, rules written by the users, monitoring by the users, graduated penalties.**

The tragedy is a warning, not a prophecy.

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Working through it

The gap this fills. The four fixes so far are either governmental (cap, price, permit) or proprietary (own). Elinor Ostrom's work documents a third route: communities that govern their own commons, sometimes for centuries, without either a state or a private owner.

The cases. Swiss alpine pastures under village grazing rules dating from the Middle Ages. Maine lobster harbours with locally defined territories and self-enforced limits. Spanish *huertas*, where irrigation water has been allocated by farmer-run courts for a thousand years.

What the survivors share — Ostrom's design principles, and each one does identifiable work:

- **Clear boundaries**, so everyone knows who is a user and what is being managed.
- **Rules written by the users**, who know the resource, rather than by a distant authority who does not.
- **Monitoring by the users**, who have both the information and the incentive to watch.
- **Graduated penalties**, starting small — which keeps enforcement credible without destroying the community that must keep living together.

The conclusion, and Ostrom's Nobel citation in 2009 rests on it. Hardin's tragedy is a warning about *unmanaged* commons, not a prophecy about all of them. Communities can and do change their own arithmetic.

The matrix, settled

The matrix, settled		
Excludable?	Rival in use?	
	yes	no
	Private goods a burrito	Club goods a streaming subscription
no	Common resources fish in the open ocean	Public goods national defense

- **Private goods:** price is gate and meter — markets work.
- **Club goods:** gate holds; access sold by subscription.
- **Common resources:** overused — cap, price, permit, or own.
- **Public goods:** undersupplied — tax and provide, *if* benefits beat costs.

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Working through it

The map, with a verdict in every cell.

Private goods (gate, rival). The price is both gate and meter: it rations the good and pays the supplier. Markets work, and this is the cell almost all of economics is about.

Club goods (gate, not rival). The gate holds even though sharing costs nothing, so access is sold by subscription. Markets function, with the efficiency caveat that some users who value the good above zero are priced out.

Common resources (no gate, rival). Overused, because nobody is charged for using them up. Repairs: cap, price, permit, own — each re-attaching the missing cost.

Public goods (no gate, not rival). Undersupplied, because nobody can charge for supplying them. Repair: tax and provide — *if* an honest cost-benefit test says the thing is worth building, and that qualification is not decoration.

What you can now do. Given any good you have never met, ask two questions, land it in a cell, and predict in advance whether a market will serve it, and if not, in which direction it will fail and which family of repair applies.